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Manifolds and coordinates

Our discussion of special relativity has led us to model the physical world as a four-dimensional continuum, called spacetime, with a Minkowski geometry. This is an example of a *manifold*. As we shall see, the more complicated spacetime geometries of general relativity are also examples of manifolds. It is therefore worthwhile discussing manifolds in general. In the following we consider general properties of manifolds commonly encountered in physics, and we concentrate in particular on Riemannian manifolds, which will be central to our discussion of general relativity.

2.1 The concept of a manifold

In general, a manifold is any set that can be continuously parameterised. The number of independent parameters required to specify any point in the set uniquely is the *dimension* of the manifold, and the parameters themselves are the *coordinates* of the manifold. An abstract example is the set of all rigid rotations of Cartesian coordinate systems in three-dimensional Euclidean space, which can be parameterised by the Euler angles. So the set of rotations is a three-dimensional manifold: each point is a particular rotation, and the coordinates of the point are the three Euler angles. Similarly, the phase space of a particle in classical mechanics can be parameterised by three position coordinates (q_1, q_2, q_3) and three momentum coordinates (p_1, p_2, p_3) , and thus the set of points in this phase space forms a six-dimensional manifold. In fact, one can regard ‘manifold’ as just a fancy word for ‘space’ in the general mathematical sense.

In its most primitive form a general manifold is simply an amorphous collection of points. Most manifolds used in physics, however, are ‘differential manifolds’, which are continuous and differentiable in the following way. A manifold is continuous if, in the neighbourhood of every point P , there are other points whose coordinates differ infinitesimally from those of P . A manifold is differentiable if it is possible to define a scalar field at each point of the manifold that can be differentiated everywhere. Both our examples above are differential manifolds.

The association of points with the values of their parameters can be thought of as a mapping of the points of a manifold into points of the Euclidean space of the same dimension. This means that ‘locally’ a manifold looks like the corresponding Euclidean space: it is ‘smooth’ and has a certain number of dimensions.

2.2 Coordinates

An N -dimensional manifold $\mathcal{M}$ of points is one for which N independent real coordinates $(x^1, x^2, \dots, x^N)$ are required to specify any point completely.¹ These N coordinates are entirely general and are denoted collectively by x^a , where it is understood that $a = 1, 2, \dots, N$.

As a technical point, we should mention that in general it may not be possible to cover the whole manifold with only one *non-degenerate* coordinate system, namely, one which ascribes a *unique* set of N coordinate values to each point, so that the correspondence between points and sets of coordinate values (labels) is one-to-one. Let us consider, for example, the points that constitute a plane. These points clearly form a two-dimensional manifold (called R^2). An example of a degenerate coordinate system on this manifold is the polar coordinates (r, ϕ) in the plane, which have a degeneracy at the origin because ϕ is indeterminate there. For this manifold, we could avoid the degeneracy at the origin by using, for example, Cartesian coordinates. For a general manifold, however, we might have no choice in the matter and might have to work with coordinate systems that cover only a portion of the manifold, called *coordinate patches*. For example, the set of points making up the surface of a sphere forms a two-dimensional manifold (called S^2). This manifold is usually ‘parameterised’ by the coordinates θ and ϕ , but ϕ is degenerate at the poles. In this case, however, it can be shown that there is no coordinate system that covers the whole of S^2 without degeneracy; the smallest number of patches needed is two. In general, a set of coordinate patches that covers the whole manifold is called an *atlas*.

Thus, in general, we do not require the whole of a manifold $\mathcal{M}$ to be covered by a single coordinate system. Instead, we may have a collection of coordinate systems, each covering some part of $\mathcal{M}$, and all these are on an equal footing. We do not regard any one coordinate system as in some way preferred.

2.3 Curves and surfaces

Given a manifold, we shall be concerned with points in it and with subsets of points that define *curves* and *surfaces*. We shall frequently define these curves

¹ The reason why the coordinates are written with superscripts rather than subscripts will become clear later.

and surfaces *parametrically*. Thus, since a curve has one degree of freedom, it depends on one parameter and so we define a curve in the manifold by the parametric equations

$$x^a = x^a(u) \quad (a = 1, 2, \dots, N),$$

where u is some parameter and $x^1(u), x^2(u), \dots, x^N(u)$ denote N functions of u .

Similarly, since a *submanifold* or *surface* of M dimensions ($M < N$) has M degrees of freedom, it depends on M parameters and is given by the N parametric equations

$$x^a = x^a(u^1, u^2, \dots, u^M) \quad (a = 1, 2, \dots, N). \quad (2.1)$$

If, in particular, $M = N - 1$ then the submanifold is called a *hypersurface*. In this case, the $N - 1$ parameters can be eliminated from these N equations to give one equation relating the coordinates, i.e.

$$f(x^1, x^2, \dots, x^N) = 0.$$

From a different but equivalent point of view, a point in a manifold is characterised by N coordinates. If the point is restricted to lie in a particular hypersurface, i.e. an $(N - 1)$ -dimensional subspace, then the point's coordinates must satisfy *one* constraint equation, namely

$$f(x^1, x^2, \dots, x^N) = 0.$$

Similarly, points in an M -dimensional subspace ($M < N$) must satisfy $N - M$ constraints

$$\begin{aligned} f_1(x^1, x^2, \dots, x^N) &= 0, \\ f_2(x^1, x^2, \dots, x^N) &= 0, \\ &\vdots \\ f_{N-M}(x^1, x^2, \dots, x^N) &= 0, \end{aligned}$$

which is an alternative to the parametric representation (2.1).

2.4 Coordinate transformations

To locate a point in a manifold we use a system of N coordinates, but the choice of these coordinates is *arbitrary*. The important idea is not the 'labels' but the points themselves and the geometrical and topological relationships between them.

We may relabel the points of a manifold by performing a *coordinate transformation* $x^a \rightarrow x'^a$ expressed by the N equations

$$\boxed{x'^a = x'^a(x^1, x^2, \dots, x^N) \quad (a = 1, 2, \dots, N),} \quad (2.2)$$

giving each new coordinate as a function of the old coordinates. Hence we view a coordinate transformation *passively* as assigning the new primed coordinates $(x'^1, x'^2, \dots, x'^N)$ to a point of the manifold whose old coordinates are $(x^1, x^2, \dots, x^N)$.

We will assume that the functions involved in (2.2) are single-valued, continuous and differentiable over the valid ranges of their arguments. Thus by differentiating each equation in (2.2) with respect to each of the old coordinates x^b we obtain the $N \times N$ partial derivatives $\partial x'^a / \partial x^b$. These may be assembled into the $N \times N$ transformation matrix²

$$\left[\frac{\partial x'^a}{\partial x^b} \right] = \begin{pmatrix} \frac{\partial x'^1}{\partial x^1} & \frac{\partial x'^1}{\partial x^2} & \cdots & \frac{\partial x'^1}{\partial x^N} \\ \frac{\partial x'^2}{\partial x^1} & \frac{\partial x'^2}{\partial x^2} & \cdots & \frac{\partial x'^2}{\partial x^N} \\ \vdots & \vdots & & \vdots \\ \frac{\partial x'^N}{\partial x^1} & \frac{\partial x'^N}{\partial x^2} & \cdots & \frac{\partial x'^N}{\partial x^N} \end{pmatrix},$$

so that rows are labelled by the index in the numerator of the partial derivative and columns by the index in the denominator. The elements of the transformation matrix are functions of the coordinates, and so the numerical values of the matrix elements are in general different when evaluated at different points in the manifold. The determinant of the transformation matrix is called the *Jacobian* of the transformation and is denoted by

$$J = \det \left[\frac{\partial x'^a}{\partial x^b} \right].$$

Clearly, the numerical value of J also varies from point to point in the manifold.

If $J \neq 0$ for some range of the coordinates x^b then it follows that in this region we can (in principle) solve the equations (2.2) for the old coordinates x^b and obtain the *inverse transformation* equations

$$x^a = x^a(x'^1, x'^2, \dots, x'^N) \quad (a = 1, 2, \dots, N).$$

² In general the notation $[]$ denotes the matrix containing the elements within the square brackets.

In a similar manner to the above, we define the inverse transformation matrix $[\partial x^a / \partial x'^b]$ and the Jacobian of the inverse transformation $J' = \det[\partial x^a / \partial x'^b]$.

Using the chain rule, it is easy to show that the inverse transformation matrix is the inverse of the transformation matrix, since

$$\sum_{b=1}^N \frac{\partial x'^a}{\partial x^b} \frac{\partial x^b}{\partial x'^c} = \frac{\partial x'^a}{\partial x'^c} = \delta_c^a = \begin{cases} 1 & \text{if } a = c, \\ 0 & \text{if } a \neq c, \end{cases}$$

where we have defined the *Kronecker delta* δ_c^a and used the fact that

$$\frac{\partial x'^a}{\partial x'^c} = \frac{\partial x^a}{\partial x^c} = 0 \quad \text{if } a \neq c,$$

because the coordinates in either the unprimed or the primed set are independent. Since the two transformation matrices are inverses of one another, it follows that $J' = 1/J$.

If we consider neighbouring points P and Q in the manifold, with coordinates x^a and $x^a + dx^a$ respectively, then in the new, primed, coordinate system the infinitesimal coordinate separation between P and Q is given by

$$dx'^a = \frac{\partial x'^a}{\partial x^1} dx^1 + \frac{\partial x'^a}{\partial x^2} dx^2 + \cdots + \frac{\partial x'^a}{\partial x^N} dx^N,$$

where it is understood that the partial derivatives on the right-hand side are evaluated at the point P . We can write this more economically as

$$dx'^a = \sum_{b=1}^N \frac{\partial x'^a}{\partial x^b} dx^b. \quad (2.3)$$

2.5 Summation convention

Our notation can be made more economical still by adopting Einstein's *summation convention*: whenever an index occurs twice in an expression, *once as a subscript and once as a superscript*, this is understood to imply a summation over the index from 1 to N , the dimension of the manifold.

Thus we can write (2.3) simply as

$$dx'^a = \frac{\partial x'^a}{\partial x^b} dx^b,$$

where, once again, it is understood that all the partial derivatives are evaluated at P . The index a appearing on each side of this equation is said to be a *free* index and may take on separately any value from 1 to N . We consider a superscript that

appears in the denominator of a partial derivative as a subscript (and vice versa). Thus the index b on the right-hand side in effect appears once as a subscript and once as a superscript, and hence there is an implied summation from 1 to N . An index that is summed over in this way is called a *dummy* index, because it can be replaced by any other index not already in use. For example, we may write

$$\frac{\partial x'^a}{\partial x^b} dx^b = \frac{\partial x'^a}{\partial x^c} dx^c,$$

since c was not already in use in the expression.

Note that the proper use of the summation convention requires that, in any term, an index should not occur more than twice and that any repeated index must occur once as a subscript and once as a superscript.

2.6 Geometry of manifolds

So far, we have considered manifolds only in a very primitive form. We have assumed that the manifold is continuous and differentiable, but aside from these properties it remains an amorphous collection of points. We have not yet defined its geometry.

Consider two infinitesimally separated points P and Q in the manifold, with coordinates x^a and $x^a + dx^a$ respectively ($a = 1, 2, \dots, N$). The *local geometry* of the manifold at the point P is determined by defining the invariant ‘distance’ or ‘interval’ ds between P and Q . In general, the distance between the points can be assigned to be any reasonably well-behaved function of the coordinates and their differentials, i.e.³

$$ds^2 = f(x^a, dx^a).$$

Clearly this function contains information on both the local geometry of the manifold at P and our chosen coordinate system. *It is the assignment at each point in the manifold of a distance between points with infinitesimally different values of the coordinates that determines the local geometry of the manifold.* To choose an example at random, a two-dimensional manifold, beloved of differential geometers for its richness, is the Finsler geometry, in which one may define coordinates ξ and ζ such that

$$ds^2 = (d\xi^4 + d\zeta^4)^{1/2}.$$

³ It is conventional to give the expression for ds^2 rather than ds .

2.7 Riemannian geometry

For developing general relativity, we are not interested in the most general geometries and can confine our attention to manifolds in which the interval is given by an expression of the form⁴ (assuming the summation convention)

$$ds^2 = g_{ab}(x) dx^a dx^b. \quad (2.4)$$

Thus, such an interval is *quadratic* in the coordinate differentials. We shall see below that the $g_{ab}(x)$ are the components of the *metric tensor* field in our chosen coordinate system. For the moment, however, we can consider them simply as a set of functions of the coordinates that determine the local geometry of the manifold at any point. Manifolds with a geometry expressible in the form (2.4) are called *Riemannian manifolds*. Strictly speaking, the manifold is only Riemannian if $ds^2 > 0$ always. If ds^2 can be positive or negative (or zero), as is the case in special relativity and general relativity, then the manifold should properly be called *pseudo-Riemannian* but is usually simply referred to as Riemannian.

The metric functions $g_{ab}(x)$ can be considered as the elements of a position-dependent $N \times N$ matrix. The metric functions can always be chosen so that $g_{ab}(x) = g_{ba}(x)$, i.e the matrix is symmetric. Suppose for argument's sake that the functions g_{ab} were not symmetric in a and b . Then we could always decompose the metric function into parts that are symmetric and antisymmetric respectively in a and b , i.e.

$$g_{ab}(x) = \frac{1}{2}[g_{ab}(x) + g_{ba}(x)] + \frac{1}{2}[g_{ab}(x) - g_{ba}(x)].$$

The contribution to ds^2 from the antisymmetric part would be $\frac{1}{2}[g_{ab}(x) - g_{ba}(x)] dx^a dx^b$, which vanishes identically, as is easily confirmed on swapping indices in one of the terms, so that any antisymmetric part of g_{ab} can safely be neglected. Thus in an N -dimensional Riemannian manifold there are $\frac{1}{2}N(N+1)$ independent metric functions $g_{ab}(x)$.

It is important to remember that the form of the metric functions can always be changed by making a change of coordinates. Since the interval between two points in the manifold is invariant under a coordinate transformation, using (2.4) and (2.3) we have

$$\begin{aligned} ds^2 &= g_{ab}(x) dx^a dx^b \\ &= g_{ab}(x) \frac{\partial x^a}{\partial x'^c} \frac{\partial x^b}{\partial x'^d} dx'^c dx'^d \\ &= g'_{cd}(x') dx'^c dx'^d, \end{aligned} \quad (2.5)$$

⁴ As we shall see in Chapter 7, this is a consequence of the equivalence principle.

where the new metric functions $g'_{ab}(x')$ in the primed coordinate system are related to those in the unprimed coordinate system by

$$g'_{cd}(x') = g_{ab}(x(x')) \frac{\partial x^a}{\partial x'^c} \frac{\partial x^b}{\partial x'^d}.$$

Clearly, the metric functions $g'_{ab}(x')$ describe the same local geometry of the manifold as do the functions $g_{ab}(x)$.

Since there are N arbitrary coordinate transformations there are really only $\frac{1}{2}N(N+1) - N = \frac{1}{2}N(N-1)$ independent degrees of freedom associated with the $g_{ab}(x)$.

2.8 Intrinsic and extrinsic geometry

It is important to realise that the local geometry or *curvature* characterised by (2.4) is an *intrinsic* property of the manifold itself, i.e. it is *independent* of whether the manifold is *embedded* in some higher-dimensional space.

It is, of course, difficult (or impossible) to imagine higher-dimensional curved manifolds, so it is instructive to consider two-dimensional Riemannian manifolds, which can often be visualised as a surface embedded in a three-dimensional Euclidean space. It is important to make a distinction, however, between the *extrinsic* properties of the surface, which are dependent on how it is embedded into a higher-dimensional space, and properties that are intrinsic to the surface itself.

This distinction is traditionally made clear by considering the viewpoint of some two-dimensional being (called a ‘bug’) confined exclusively to the two-dimensional surface. Such a being would believe that it is able to look and measure in all directions, whereas it is in fact limited to making measurements of distance, angle etc. only within the surface. For example, it would receive light signals that had travelled within the two-dimensional surface. Properties of the geometry that are accessible to the bug are called *intrinsic*, whereas those that depend on the viewpoint of a higher-dimensional creature (who is able to see how the surface is shaped in the three-dimensional space) are called *extrinsic*.

The bug is able to define a coordinate system and measure distances in the surface (e.g. by counting how many steps it has to take) from one point to another. It can thus define a set of metric functions $g_{ab}(x)$ that characterise the intrinsic geometry of the surface (as expressed in the bug’s chosen coordinate system).

Consider, for example, a two-dimensional plane surface, such as a flat sheet of paper, in our three-dimensional Euclidean space. The bug can label the entire sheet using rectangular Cartesian coordinates, so that the distance ds measured

over the surface between any pair of points whose coordinate separations are dx and dy is given by

$$ds^2 = dx^2 + dy^2.$$

If this sheet is then rolled up into a cylinder, the bug would not be able to detect any differences in the geometrical properties of the surface (see Figure 2.1).

To the bug, the angles of a triangle still add up to 180° , the circumference of a circle is still $2\pi r$ etc. The proof of this fact is simple – the surface can simply be *unrolled* back to a flat surface without buckling, tearing or otherwise distorting it. A more mathematical approach is to note that if one parameterises the surface of the cylinder (of radius a) using cylindrical coordinates (z, ϕ) , the distance ds measured over the surface between any two points whose coordinate separations are dz and $d\phi$ is given by

$$ds^2 = dz^2 + a^2 d\phi^2.$$

By making the simple change of variables $x = z$ and $y = a\phi$ we recover the expression $ds^2 = dx^2 + dy^2$, which is valid over the whole surface, and so the intrinsic geometry is identical to that of a flat plane. Thus the surface of a cylinder is not intrinsically curved; its curvature is extrinsic and a result of the way it is embedded in three-dimensional space. Even if one were to crumple up the sheet of paper (without tearing it), so that its extrinsic geometry in three-dimensional space was very complicated, its intrinsic geometry would still be that of a plane.

The situation is somewhat different for a 2-sphere, i.e. a spherical surface, embedded in three-dimensional Euclidean space. Once again the surface is manifestly curved extrinsically on account of its embedding. Additionally, however, it cannot be formed from a flat sheet of paper without tearing or deformation. Its intrinsic geometry – based on measurements within the surface – differs from the intrinsic (Euclidean) geometry of the plane. This problem is well known to

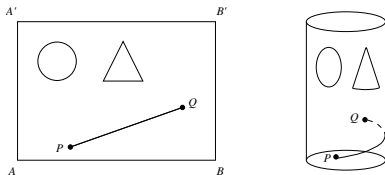


Figure 2.1 Rolling up a flat sheet of paper into a cylinder.

cartographers. Mathematically, if we parameterise a sphere (of radius a) by the usual angular coordinates (θ, ϕ) then

$$ds^2 = a^2(d\theta^2 + \sin^2 \theta d\phi^2),$$

which cannot be transformed to the Euclidean form $ds^2 = dx^2 + dy^2$ over the whole surface by any coordinate transformation. Thus the surface of a sphere is intrinsically curved.

We note, however, that *locally* at any point A on the spherical surface we can define a set of Cartesian coordinates, so that $ds^2 = dx^2 + dy^2$ is valid in the neighbourhood of A . For example, the street layout of a town can be accurately represented by a flat map, whereas the entire globe can only be represented by performing projections that distort distance and/or angles. As an idea of what can happen to local Cartesian coordinate systems far from the point A where they are defined, consider Figure 2.2. If a bug starts at A and travels in the locally defined x -direction to B , it observes that C still lies in the y -direction. If instead the bug travels from A to C , it finds that B still lies in the x -direction. The non-Euclidean geometry of the spherical surface is also apparent from the fact that the angles of the triangle ABC sum to 270° .

We may take our discussion one step further, dispense with the three-dimensional space and embedding-related extrinsic geometry and consider the surfaces in isolation. Intrinsic geometry is all that remains with any meaning. For example, when we talk of the curvature of spacetime in general relativity, we must resist any temptation to think of spacetime as embedded in any ‘higher’ space. Any such embedding, whether or not it is physically realised, would be irrelevant to our discussion. Nevertheless, in developing our intuition for

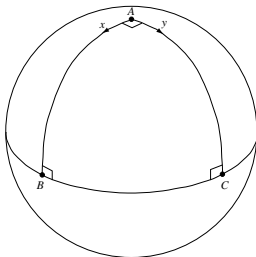


Figure 2.2 A two-dimensional spherical surface.

curved manifolds it often remains useful to imagine two-dimensional surfaces embedded in three-dimensional Euclidean space.

2.9 Examples of non-Euclidean geometry

Let us develop our intuition for non-Euclidean geometry by considering in more detail the surface of a sphere. We begin by imagining the usual Cartesian coordinate system (x, y, z) defining a Euclidean three-dimensional space with line element

$$ds^2 = dx^2 + dy^2 + dz^2. \quad (2.6)$$

Now, suppose that we have a sphere of radius a with its centre at the origin of our coordinate system. We will now ask the following question: what is the line element on the surface of the sphere?

The equation defining the sphere is

$$x^2 + y^2 + z^2 = a^2.$$

So, differentiating this equation, we obtain

$$2x dx + 2y dy + 2z dz = 0,$$

and we can write an equation for dz ,

$$dz = -\frac{x dx + y dy}{z} = \frac{-(x dx + y dy)}{[a^2 - (x^2 + y^2)]^{1/2}}. \quad (2.7)$$

Thus, equation (2.9) provides a constraint on dz that keeps us on the surface of the sphere if we are displaced by small amounts dx and dy from an arbitrary point on the sphere (for example, the point A in Figure 2.2). Substituting for dz in (2.6) gives us the interval for such constrained displacements:

$$ds^2 = dx^2 + dy^2 + \frac{(x dx + y dy)^2}{a^2 - (x^2 + y^2)}, \quad (2.8)$$

which is the line element for the surface of the sphere in terms of our chosen coordinates (as shown in Figure 2.2), taking A as the origin of x and y . We see that this line element reduces to the Euclidean form $ds^2 = dx^2 + dy^2$ in the neighbourhood of A . Practically, one could construct the coordinate curves $x = \text{constant}$ and $y = \text{constant}$ on the surface of the sphere by creating a standard (x, y) coordinate grid in the tangent plane at A and ‘projecting’ vertically down onto the spherical surface.

We may obtain an alternative form for the line element by making the substitutions

$$x = \rho \cos \phi, \quad y = \rho \sin \phi,$$

and after a little algebra we obtain⁵

$$ds^2 = \frac{a^2 d\rho^2}{a^2 - \rho^2} + \rho^2 d\phi^2. \quad (2.9)$$

As above, one can construct the ρ and ϕ coordinate curves on the sphere by creating a standard (ρ, ϕ) coordinate system in the tangent plane at A and projecting vertically down onto the surface. We also note that this line element contains a ‘hidden symmetry’, namely our freedom to choose an arbitrary point on the sphere as the origin $\rho = 0$.

The observant reader will have noticed that the line elements (2.8) and (2.9) have singularities at $\sqrt{x^2 + y^2} = a$, or, equivalently, $\rho = a$, corresponding to the equator of the sphere (relative to A). From our embedding picture, it is clear why the (x, y) and (ρ, ϕ) coordinates cover the surface of the sphere uniquely only up to this point. We note, however, that there is nothing pathological in the intrinsic geometry of the 2-sphere at the equator. What we have observed is only a *coordinate singularity*, which has resulted simply from choosing coordinates with a restricted domain of validity. Although the embedding picture we have adopted gives both the (x, y) and (ρ, ϕ) coordinate systems a clear geometrical meaning in our three-dimensional Euclidean space, it is important to realise that a bug confined to the two-dimensional surface of the sphere could, if it wished, have defined these coordinate systems to describe the intrinsic geometry without any reference to an embedding in higher dimensions.

We can make an analogous construction to find the metric for a 3-sphere embedded in *four-dimensional* Euclidean space. The metric for the four-dimensional Euclidean space is

$$ds^2 = dx^2 + dy^2 + dz^2 + dw^2, \quad (2.10)$$

and, by analogy with the example above, the equation defining a 3-sphere is

$$x^2 + y^2 + z^2 + w^2 = a^2.$$

Differentiating as before gives

$$2x dx + 2y dy + 2z dz + 2w dw = 0,$$

⁵ Note that the line elements (2.8) and (2.9) look different from the metric we would write down using standard spherical polars, $ds^2 = a^2 d\theta^2 + a^2 \sin^2 \theta d\phi^2$. Nonetheless, both are valid line elements for the two-dimensional surface of a sphere.

and so substituting for dw in (2.10) gives the line element:

$$ds^2 = dx^2 + dy^2 + dz^2 + \frac{(x dx + y dy + z dz)^2}{a^2 - (x^2 + y^2 + z^2)}.$$

Transforming to spherical polar coordinates

$$x = r \sin \theta \cos \phi,$$

$$y = r \sin \theta \sin \phi,$$

$$z = r \cos \theta,$$

we obtain an alternative form for the line element:

$$ds^2 = \frac{a^2}{a^2 - r^2} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2. \quad (2.11)$$

Notice that, in the limit $a \rightarrow \infty$, the metric tends to the form

$$ds^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2,$$

which is simply the metric of ordinary Euclidean three-dimensional space $ds^2 = dx^2 + dy^2 + dz^2$, rewritten in spherical polar coordinates. The line element (2.11) therefore describes a *non-Euclidean* three-dimensional space. We note that this line element also has a singularity, this time at $r = a$. As one might expect from our discussion above, this is once again just a coordinate singularity, although our existence as three-dimensional ‘bugs’ makes the geometric reason for this less straightforward to visualise!

2.10 Lengths, areas and volumes

For a given set of metric functions $g_{ab}(x)$, (2.4), it is useful to know how to compute the lengths of curves and the ‘areas’ and ‘volumes’ of subregions of the manifold.

The lengths of curves follow immediately from the line element. Suppose that the points A and B are joined by some path; then the length of this curve is given by

$$L_{AB} = \int_A^B ds = \int_A^B |g_{ab} dx^a dx^b|^{1/2},$$

where the integral is evaluated along the curve. As indicated, the absolute value of ds is taken before the square root is evaluated when considering

pseudo-Riemannian manifolds. If the equation of the curve $x^a(u)$ is given in terms of some parameter u then

$$L_{AB} = \int_{u_A}^{u_B} \left| g_{ab} \frac{dx^a}{du} \frac{dx^b}{du} \right|^{1/2} du, \quad (2.12)$$

where u_A and u_B are the values of the parameter u at the endpoints of the curve.

For the calculation of areas and volumes, let us begin by considering the simple case where the metric is diagonal, i.e. $g_{ab}(x) = 0$ for $a \neq b$.⁶ In this case the line element takes the form

$$ds^2 = g_{11}(dx^1)^2 + g_{22}(dx^2)^2 + \cdots + g_{NN}(dx^N)^2. \quad (2.13)$$

Such a system of coordinates is called *orthogonal* since, at all points in the manifold, any pair of coordinate curves cross at right angles, as is clear from (2.13). Thus, in orthogonal coordinate systems the ideas of area and volume can be built up simply. Consider, for example, an element of area in the (x^1, x^2) -surface defined by $x^a = \text{constant}$ for $a = 3, 4, \dots, N$. Suppose that the area element is defined by the *coordinate* lengths dx^1 and dx^2 (see Figure 2.3). The *proper lengths* of the two line segments will be $\sqrt{g_{11}} dx^1$ and $\sqrt{g_{22}} dx^2$ respectively. Thus the element of area is⁷

$$dA = \sqrt{|g_{11}g_{22}|} dx^1 dx^2. \quad (2.14)$$

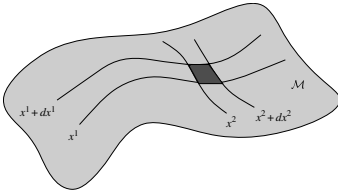


Figure 2.3 An element of area, on a manifold $\mathcal{M}$, defined by the coordinate intervals dx^1 and dx^2 . The proper lengths dl^1 and dl^2 of these intervals are related to dx^1 and dx^2 by the metric functions. If the coordinate lines are orthogonal then the area of is $dl^1 dl^2$.

⁶ The general case is discussed in Section 2.14.

⁷ We have implicitly assumed here that the manifold is strictly Riemannian. If the manifold is pseudo-Riemannian, some of the elements g_{ab} in (2.13) may be negative (see Section 2.13), and then we require the modulus signs.

Similarly, for 3-volumes in the (x^1, x^2, x^3) -surface defined by $x^a = \text{constant}$ for $a = 4, 5, \dots, N$, we have

$$d^3V = \sqrt{|g_{11}g_{22}g_{33}|} dx^1 dx^2 dx^3. \quad (2.15)$$

We may, of course, define 3-volumes for any other three-dimensional subspace. We can define higher-dimensional ‘volume’ elements in a similar way until we reach the N -dimensional volume element

$$d^N V = \sqrt{|g_{11}g_{22} \cdots g_{NN}|} dx^1 dx^2 \cdots dx^N.$$

As examples of working with such metric functions, let us consider the non-Euclidean spaces discussed in Section 2.9. We begin with the line element (2.9),

$$ds^2 = \frac{a^2 d\rho^2}{a^2 - \rho^2} + \rho^2 d\phi^2, \quad (2.16)$$

which describes two-dimensional geometry on the surface of a sphere in terms of the coordinates (ρ, ϕ) , the geometrical meanings of which are illustrated in Figure 2.4 assuming an embedding in three-dimensional Euclidean space. From (2.16) we see that this coordinate system is orthogonal, with $g_{\rho\rho} = a^2/(a^2 - \rho^2)$ and $g_{\phi\phi} = \rho^2$ (no sums on ρ or ϕ).⁸ Let us consider a circle defined by $\rho = R$,

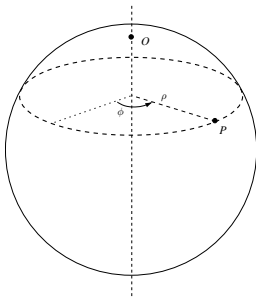


Figure 2.4 The surface of a sphere parameterised by the coordinates (ρ, ϕ) appearing in the line element (2.16).

⁸ This form of notation is quite common, once a particular coordinate system has been chosen, and it is usually clear from the context that no summation is implied.

where R is some constant, and calculate its length, its area and the distance from its centre to the perimeter.

From (2.12) and (2.16), the distance *in the surface* from the centre to the perimeter along a line of constant ϕ is

$$D = \int_0^R \frac{a}{(a^2 - \rho^2)^{1/2}} d\rho = a \sin^{-1} \left(\frac{R}{a} \right),$$

while the circumference of the circle is given by

$$C = \int_0^{2\pi} R d\phi = 2\pi R.$$

Similarly, from (2.14) we have, for the area of the spherical surface enclosed by C ,

$$A = \int_0^{2\pi} \int_0^R \frac{a}{(a^2 - \rho^2)^{1/2}} \rho d\rho d\phi = 2\pi a^2 \left[1 - \left(1 - \frac{R^2}{a^2} \right)^{1/2} \right].$$

Note that if we rewrite the circumference C and area A in terms of the distance D then we obtain

$$C = 2\pi a \sin \left(\frac{D}{a} \right) \quad \text{and} \quad A = 2\pi a^2 \left[1 - \cos \left(\frac{D}{a} \right) \right]. \quad (2.17)$$

Thus, as D increases, both the circumference and area of the circle increase *until the point when $D = \pi a/2$, after which both C and A become smaller as D increases.*

In fact there is a slight subtlety here. As noted earlier, if we attempt to parameterise points beyond the equator of the sphere using the coordinates (ρ, ϕ) , the system becomes degenerate, i.e. there is more than one point in the surface with the same coordinates. The degenerate nature of the (ρ, ϕ) coordinate system means that some care is required, for example, in calculating the total area of the surface. By symmetry this is given by

$$A_{\text{tot}} = 2 \int_0^{2\pi} \int_0^a \frac{a}{(a^2 - \rho^2)^{1/2}} \rho d\rho d\phi = 4\pi a^2.$$

Although we cannot easily visualise the geometry, we can perform similar calculations for the line element (2.11),

$$ds^2 = \frac{a^2}{a^2 - r^2} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2, \quad (2.18)$$

which describes a non-Euclidean three-dimensional space that tends to Euclidean three-dimensional space as $a \rightarrow \infty$. Let us consider a 2-sphere of coordinate radius $r = R$ and calculate the circumference around the equator, the area, the volume and the distance from its centre to the surface of the sphere.

From (2.12) and (2.18), the distance from the centre to the surface along a line $\theta = \text{constant}$, $\phi = \text{constant}$ is

$$D = \int_0^R \frac{a \, dr}{(a^2 - r^2)^{1/2}} = a \sin^{-1} \left(\frac{R}{a} \right).$$

Noting that the equator of the sphere is the curve $r = R$, $\theta = \pi/2$, its circumference is $C = \int_0^{2\pi} R \, d\phi = 2\pi R$, while the area of the surface $r = R$ and the volume it encloses are obtained from (2.14) and (2.15) and read

$$\begin{aligned} A &= \int_0^{2\pi} \int_0^\pi R^2 \sin \theta \, d\theta \, d\phi = 4\pi R^2, \\ V &= \int_0^{2\pi} \int_0^\pi \int_0^R \frac{ar^2 \sin \theta}{(a^2 - r^2)^{1/2}} \, dr \, d\theta \, d\phi \\ &= 4\pi a^3 \left\{ \frac{1}{2} \sin^{-1} \left(\frac{R}{a} \right) + \frac{R}{a} \left[1 - \left(\frac{R}{a} \right)^2 \right]^{1/2} \right\}. \end{aligned}$$

It is not difficult to see that the familiar results of three-dimensional Euclidean space are recovered when $R/a \ll 1$. Once again, we can rewrite our results in terms of D rather than R , and we find that C , A and V all have maximum values at $D = \pi a/2$. By analogy with the above two-dimensional example, the total volume of this space is

$$V_{\text{tot}} = 2 \int_0^{2\pi} \int_0^\pi \int_0^a \frac{ar^2 \sin \theta}{(a^2 - r^2)^{1/2}} \, dr \, d\theta \, d\phi = 2\pi^2 a^3.$$

The three-dimensional non-Euclidean space described by the line element (2.18) thus has a *finite* volume. We can generate a line element for an *infinite* non-Euclidean three-dimensional space by making the substitution $a = ib$, i.e. choosing the ‘radius’ of the space to be pure imaginary. The line element (2.18) then becomes

$$ds^2 = \frac{b^2}{b^2 + r^2} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta \, d\phi^2.$$

If we again consider the sphere defined by $r = R$, we find easily that in this space $C = 2\pi R$ and $A = 4\pi R^2$ as before but the distance from the centre of the sphere to its surface is now given by $D = b \sinh^{-1}(R/b)$. In this case, one finds that C , A and the volume V of the sphere are all monotonically increasing functions.

2.11 Local Cartesian coordinates

We now introduce a key property of Riemannian manifolds, to which we have alluded in earlier sections. For the moment we will confine our attention to

manifolds that are strictly Riemannian, so that $ds^2 > 0$ always, but subsequently we will extend our discussion to pseudo-Riemannian spaces, in which ds^2 can be of either sign (or zero).

For a general Riemannian manifold, it is *not* possible to perform a coordinate transformation $x^a \rightarrow x'^a$ that will take the line element $ds^2 = g_{ab}(x) dx^a dx^b$ into the Euclidean form

$$ds^2 = (dx'^1)^2 + (dx'^2)^2 + \cdots + (dx'^N)^2 = \delta_{ab} dx'^a dx'^b,$$

at every point in the manifold. This is clear, since there are $N(N+1)/2$ independent metric functions $g_{ab}(x)$ but only N coordinate transformation functions $x'^a(x)$. As we shall now demonstrate, however, it is always possible to make a coordinate transformation such that *in the neighbourhood of some specified point P* the line element takes the Euclidean form. In other words, we can always find coordinates x'^a such that *at the point P* the new metric functions $g'_{ab}(x')$ satisfy

$$g'_{ab}(P) = \delta_{ab}, \quad (2.19)$$

$$\left. \frac{\partial g'_{ab}}{\partial x'^c} \right|_P = 0. \quad (2.20)$$

Thus, in the neighbourhood of P , we have

$$g'_{ab}(x') = \delta_{ab} + \mathcal{O}[(x' - x'_P)^2].$$

Such coordinates are called *local Cartesian coordinates* at P .

From (2.5), the general transformation rule for the metric functions is

$$g'_{ab} = \frac{\partial x^c}{\partial x'^a} \frac{\partial x^d}{\partial x'^b} g_{cd},$$

which we require to satisfy the conditions (2.19) and (2.20) at our chosen point P . If x^a is an arbitrary given coordinate system and x'^a is the desired system then there will be some relation $x^a(x')$ connecting the two sets of coordinates. Although we do not (as yet) know the required transformation, we can define it in terms of its Taylor expansion about P :

$$\begin{aligned} x^a(x') &= x_P^a + \left(\frac{\partial x^a}{\partial x'^b} \right)_P (x'^b - x_P'^b) \\ &+ \frac{1}{2} \left(\frac{\partial^2 x^a}{\partial x'^b \partial x'^c} \right)_P (x'^b - x_P'^b) (x'^c - x_P'^c) \\ &+ \frac{1}{6} \left(\frac{\partial^3 x^a}{\partial x'^b \partial x'^c \partial x'^d} \right)_P (x'^b - x_P'^b) (x'^c - x_P'^c) (x'^d - x_P'^d) + \cdots \end{aligned}$$

The numbers of free independent variables we have for this purpose are as follows:

$$\begin{array}{ll}
 (\partial x^a / \partial x'^b)_P & \text{has } N^2 \text{ independent values,} \\
 (\partial^2 x^a / \partial x'^b \partial x'^c)_P & \text{has } \frac{1}{2} N^2 (N+1) \text{ independent values,} \\
 (\partial^3 x^a / \partial x'^b \partial x'^c \partial x'^d)_P & \text{has } \frac{1}{6} N^2 (N+1)(N+2) \text{ independent values,}
 \end{array}$$

where we have made use of the fact that the second set of quantities is symmetric in b and c and the third set of quantities is totally symmetric in b , c and d . We may compare this with the number of independent parameters we may want to fix:

$$\begin{array}{ll}
 g'_{ab}(P) & \text{has } \frac{1}{2} N(N+1) \text{ independent values,} \\
 (\partial g'_{ab} / \partial x'^c)_P & \text{has } \frac{1}{2} N^2 (N+1) \text{ independent values,} \\
 (\partial^2 g'_{ab} / \partial x'^c \partial x'^d)_P & \text{has } \frac{1}{4} N^2 (N+1)^2 \text{ independent values.}
 \end{array}$$

The first question is whether we can satisfy the requirement (2.19). This condition consists of $N(N+1)/2$ independent equations, and to satisfy them we have N^2 free values in $(\partial x^a / \partial x'^b)_P$. Therefore, they can indeed be satisfied, leaving $N(N-1)/2$ numbers to spare! These spare degrees of freedom correspond exactly to the number of independent N -dimensional ‘rotations’ that leave δ_{ab} unchanged.

The next question is whether we can satisfy the requirement (2.20). This condition consists of $N^2(N+1)/2$ independent equations, and we can choose an equal number of free values $(\partial^2 x^a / \partial x'^b \partial x'^c)_P$ to satisfy them.

The final question is whether we can continue in this way to higher orders. In other words, can we find a set of coordinates x'^a such that $(\partial^2 g'_{ab} / \partial x'^c \partial x'^d)_P = 0$? This condition consists of $N^2(N+1)^2/4$ independent equations, but we have only $N^2(N+1)(N+2)/6$ free values in $(\partial^2 g'_{ab} / \partial x'^c \partial x'^d)_P$, so these equations cannot in general be satisfied. This means that there are $N^2(N^2-1)/12$ ‘degrees of freedom’ among the second derivatives $(\partial^2 g'_{ab} / \partial x'^c \partial x'^d)_P$, i.e. in general at least this number of second derivatives will not vanish.

Although we have shown, in principle, that it is always possible to define local Cartesian coordinates at any given point P , we have not shown explicitly how to find such coordinates. We will return to this point in Chapter 3.

2.12 Tangent spaces to manifolds

To aid our intuition of local Cartesian coordinates, it is useful to consider the simple example of a two-dimensional Riemannian manifold, which we can often

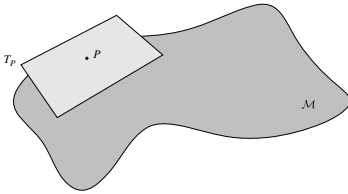


Figure 2.5 The tangent plane T_P to the curved surface $\mathcal{M}$ at the point P .

consider as a generally curved surface embedded in three-dimensional Euclidean space. A simple example is the surface of a sphere, shown in Figure 2.2. As we have shown, at any arbitrary point P we can find coordinates x and y (say) such that in the neighbourhood of P we have

$$ds^2 = dx^2 + dy^2.$$

It thus follows that a Euclidean two-dimensional space (a plane) will match the manifold locally at P . This Euclidean space is called the *tangent space* T_P to the manifold at P . In other words, in terms of our embedding picture a plane can always be drawn at any arbitrary point on a two-dimensional Riemannian surface in such a way that it is locally tangential to the surface (see Figure 2.5). Although the tangent plane to a surface at P gives a useful way of visualising the tangent space of a manifold at a point, this view can be misleading. As we stressed earlier, a manifold should be regarded as an entity in itself: there is no need for a higher-dimensional space in which it and its tangent spaces are embedded.

We may extend the idea of tangent spaces to higher dimensions. At an arbitrary point P in an N -dimensional Riemannian manifold we can find a coordinate system such that in the neighbourhood of P the line element is Euclidean. Thus, an N -dimensional Euclidean space matches the manifold locally at P . Just as each point P of an embedded two-dimensional surface has its tangent plane, making contact with the surface at P , so each point P of a manifold has a tangent space T_P attached to it.

2.13 Pseudo-Riemannian manifolds

Thus far we have confined our attention almost exclusively to strictly Riemannian manifolds, in which $ds^2 > 0$ always. In a pseudo-Riemannian manifold, however, ds^2 can be either positive, negative or zero and it is therefore much

harder to visualise even two-dimensional manifolds of this type. Nevertheless, the mathematical tools we have developed so far are straightforwardly applied to pseudo-Riemannian manifolds with little modification.

The simplest way to understand pseudo-Riemannian manifolds is to consider the transformation to local 'Cartesian' coordinates at some arbitrary point P . You will notice from Section 2.11 that our argument showing that the condition (2.20) holds for the derivatives of the metric functions in a Riemannian manifold can be extended immediately to the pseudo-Riemannian case. Let us assume that the coordinate system x^a already satisfies this condition. However, the condition (2.19) on the values of the metric functions themselves requires further investigation. Let us now attempt to obtain a new coordinate system x'^a in which (2.19) is also satisfied. We note in passing that, in order for (2.20) to remain valid, the new coordinates x'^a must be related to the old ones x^a by a *linear* transformation, $x'^a = X^a_b x^b$, where the X^a_b are *constants*.

In general, at a point P the metric functions in the new coordinate system are given in terms of the original metric functions by

$$g'_{ab}(P) = \left(\frac{\partial x^c}{\partial x'^a} \right)_P \left(\frac{\partial x^d}{\partial x'^b} \right)_P g_{cd}(P). \quad (2.21)$$

Let us define symmetric matrices G and G' having elements $g_{ab}(P)$ and $g'_{ab}(P)$ respectively. Similarly, we can define a matrix X having elements $(\partial x^a / \partial x'^b)_P$. Then, in matrix notation, (2.21) can be written as

$$G' = X^T G X.$$

Since G is symmetric, it can be diagonalised by this similarity transformation, provided that we choose the columns of X to be the normalised eigenvectors of G . Then $G' = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_N)$, where λ_a is the a th eigenvalue of G (the eigenvalues must all be real).

In a strictly Riemannian manifold, $ds^2 = g_{ab} dx^a dx^b$ is always positive at any point P . Thus the matrix $G \equiv [g_{ab}]$ at any point must be positive definite, i.e. all its eigenvalues must be positive. At an arbitrary point in a pseudo-Riemannian manifold, however, ds^2 can be positive, negative or zero, depending on the direction in which one moves from P . Correspondingly, some of the eigenvalues of G are negative.

If we now scale our coordinates according to $x'^a \rightarrow x'^a / \sqrt{|\lambda_a|}$ (note that here there is no sum on a), we obtain at the point P

$$G' = \text{diag}(\pm 1, \pm 1, \dots, \pm 1),$$

where the $+$ and $-$ signs depend on whether the corresponding eigenvalue is positive or negative. Thus, at any arbitrary point P in a pseudo-Riemannian

manifold, it is always possible to find a coordinate system x'^a such that in the neighbourhood of P we have

$$g'_{ab}(x') = \eta_{ab} + \mathcal{O}[(x' - x'_P)^2],$$

where $[\eta_{ab}] = \text{diag}(\pm 1, \pm 1, \dots, \pm 1)$. The number of positive entries minus the number of negative entries in $[\eta_{ab}]$ is called the *signature* of the manifold and is the same at all points.

It follows that, at any arbitrary point P in a pseudo-Riemannian manifold, an N -dimensional space with line element

$$ds^2 = \pm(dx^1)^2 \pm (dx^2)^2 \pm \dots \pm (dx^N)^2$$

will match the manifold locally at P . Such a space is called *pseudo-Euclidean* and is the tangent space T_P to the pseudo-Riemannian manifold at P . An example of a pseudo-Euclidean space is the four-dimensional Minkowski spacetime of special relativity, which has the line element

$$ds^2 = d(ct)^2 - dx^2 - dy^2 - dz^2$$

when expressed in coordinates corresponding to a Cartesian inertial frame. Minkowski spacetime thus has a signature of -2 .

2.14 Integration over general submanifolds

In Section 2.10, we restricted our calculation of ‘volumes’ to coordinate systems x^a that were orthogonal and to submanifolds that were obtained simply by allowing some of the coordinates to be constants. In fact neither of these simplifications is necessary, and we are now in a position to consider the general case.

Let us begin by calculating the full N -dimensional volume element $d^N V$ in an N -dimensional (pseudo-)Riemannian manifold. From Section 2.10, we know that if we are working in an orthogonal coordinate system then this volume element is given by

$$d^N V = \sqrt{|g_{11}g_{22} \dots g_{NN}|} dx^1 dx^2 \dots dx^N.$$

For such a coordinate system the matrix G is given by

$$G \equiv [g_{ab}] = \text{diag}(g_{11}, g_{22}, \dots, g_{NN}),$$

so that its determinant is simply the product of the diagonal elements,

$$\det G = g_{11}g_{22} \dots g_{NN}.$$

It is usual to denote $\det G$ simply by the symbol g . Thus, we may rewrite the volume element as

$$\boxed{dx^N V = \sqrt{|g|} dx^1 dx^2 \cdots dx^N.} \quad (2.22)$$

What we will now show is that this expression remains valid for an *arbitrary* coordinate system.

The key to proving the general result (2.22) for the volume element at some arbitrary point P in the manifold is to transform to local Cartesian coordinates x'^a at P . We know that a small N -dimensional region at P will have volume $dx'^1 dx'^2 \cdots dx'^N$. In *any* other general coordinate system x^a it is a well-known result that

$$dx'^1 dx'^2 \cdots dx'^N = J dx^1 dx^2 \cdots dx^N, \quad (2.23)$$

where the Jacobian factor J is given by

$$J = \det \left[\frac{\partial x'^a}{\partial x^b} \right].$$

If, as in Section 2.13, we use X to denote the transformation matrix $[\partial x^a / \partial x'^b]$ then $J = \det(X^{-1}) = (\det X)^{-1}$. Defining matrices G and G' as those having elements $g_{ab}(P)$ and $g'_{ab}(P)$ respectively, we have (see Section 2.13)

$$G' = X^T G X. \quad (2.24)$$

Taking determinants of both sides of (2.24) and denoting $\det G$ by g and $\det G'$ by g' we obtain

$$g' = (\det X)^2 g = \frac{1}{J^2} g.$$

Since the x'^a are locally Cartesian coordinates, $G' = \text{diag}(\pm 1, \pm 1, \dots, \pm 1)$, where the number of positive and negative signs depends on the signature of the manifold. Thus we have $g' = \pm 1$, so that $g = \pm J^2$. Hence, we obtain the required result:

$$dx^N V = dx'^1 dx'^2 \cdots dx'^N = \sqrt{|g|} dx^1 dx^2 \cdots dx^N.$$

We now turn to the question how to integrate over submanifolds that are not defined simply by setting some of the coordinates x^a to be constant. Consider some M -dimensional subspace of an N -dimensional manifold. In general, the subspace can be defined by the N parametric equations

$$x^a = x^a(u^1, u^2, \dots, u^M),$$

where the u^i ($i = 1, 2, \dots, M$) may be considered simply as a set of coordinates that parameterise the subspace. If we consider two neighbouring points in the

subspace whose parameters differ by du^i then the coordinate separation between these points is simply

$$dx^a = \frac{\partial x^a}{\partial u^i} du^i.$$

Thus the distance ds between the points is given by

$$ds^2 = g_{ab} dx^a dx^b = g_{ab} \frac{\partial x^a}{\partial u^i} \frac{\partial x^b}{\partial u^j} du^i du^j,$$

which we may write as

$$ds^2 = h_{ij} du^i du^j,$$

where the h_{ij} are the *induced* metric functions on the subspace and are given by

$$h_{ij} = g_{ab} \frac{\partial x^a}{\partial u^i} \frac{\partial x^b}{\partial u^j}. \quad (2.25)$$

Thus we can now work simply in terms of this subspace and regard it as a manifold in itself. Thus the volume element for integrals over this subspace is given in terms of the parameters u^i by

$$d^M V = \sqrt{|h|} du^1 du^2 \dots du^M,$$

where $h = \det[h_{ij}]$.

It is also worth noting here that the relation (2.25) is the key to determining whether one can embed a given manifold in another manifold of higher dimension. Suppose we begin with an M -dimensional manifold possessing the metric $h_{ij}(u)$ when labelled with the coordinates $u^i (i = 1, 2, \dots, M)$. In order to embed this manifold in an N -dimensional manifold (where $N > M$) with metric $g_{ab}(x)$ in the coordinates $x^a (a = 1, 2, \dots, N)$, then one must be able to satisfy the relation (2.25).

2.15 Topology of manifolds

In this chapter we have discussed only the local geometry of manifolds, which is defined at any point by the line element (2.4) giving the distance between points with infinitesimal coordinate separations. In addition to this local geometry a manifold also has a global geometry or *topology*. The topology of a manifold is defined by identifying certain sets of points, that is, regarding them as being coincident. For example, in Figure 2.1, we identified the line AA' with the line BB' . This property can be detected by a 'bug' on the surface, since by continuing in a straight line in a certain direction, it can get back to where it started. Thus a

topology (in this case the fact that the space is periodic in one of the coordinates) is an intrinsic property of a manifold.

We shall see that general relativity is a 'local' theory, in which the local geometry (or curvature) of the four-dimensional spacetime manifold at any point is determined by the energy density of matter and/or radiation at that point. The field equations of general relativity do *not* constrain the global topology of the spacetime manifold.

Exercises

- 2.1 In three-dimensional Euclidean space R^3 , write down expressions for the change of coordinates from Cartesian coordinates $[x^a] = (x, y, z)$ to spherical polar coordinates $[x'^a] = (r, \theta, \phi)$. Obtain expressions for the transformation and inverse transformation matrices in terms of the primed coordinates. By calculating the Jacobians J and J' for the transformation and its inverse, find where the transformation is non-invertible.
- 2.2 Write down the line element for three-dimensional Euclidean space in spherical polar coordinates x^a and cylindrical polar coordinates x'^a . Hence identify the metric functions in each coordinate system and show that they obey

$$g'_{cd}(x') = g_{ab}(x) \frac{\partial x^a}{\partial x'^c} \frac{\partial x^b}{\partial x'^d}.$$

- 2.3 In three-dimensional Euclidean space a coordinate system x'^a is related to the Cartesian coordinates x^a by

$$x^1 = x'^1 + x'^2, \quad x^2 = x'^1 - x'^2, \quad x^3 = 2x'^1 x'^2 + x'^3.$$

Describe the coordinate surfaces in the primed system. Obtain the metric functions g'_{ab} in the primed system and hence show that these coordinates are not orthogonal. Calculate the volume element dV in the primed coordinate system.

- 2.4 Consider the surface of a 2-sphere embedded in three-dimensional Euclidean space. In a *stereographic projection*, one assigns coordinates (ρ, ϕ) to each point on the surface of the sphere. The ϕ -coordinate is the standard azimuthal polar angle. The ρ -coordinate of each point is obtained by drawing a straight line in three dimensions from the south pole of the sphere through the point in question and extending the line until it intersects the tangent plane to the north pole of the sphere; the ρ -coordinate is then the distance in the tangent plane from the north pole to the intersection point. Show that the line element for the surface of the sphere in these coordinates is

$$ds^2 = \frac{d\rho^2}{(1 + \rho^2/a^2)^2} + \frac{\rho^2}{1 + \rho^2/a^2} d\phi^2.$$

At what point(s) on the sphere are these coordinates degenerate? If instead one works in terms of the Cartesian coordinates x and y in the tangent plane at the north pole, what is the corresponding form of the line element? At what point(s) on the sphere are these new coordinates degenerate?

- 2.5 Consider the surface of the Earth, which we assume for simplicity to be a 2-sphere of radius a . In terms of standard polar coordinates (θ, ϕ) , the *longitude* of a point, in radians, rather than the usual degrees, is simply ϕ (measured eastwards from the Greenwich meridian), whereas its latitude $\lambda = \pi/2 - \theta$ radians. Show that the line element on the Earth's surface in these coordinates is

$$ds^2 = a^2(d\lambda^2 + \cos^2 \lambda d\phi^2).$$

To make a map of the Earth's surface, we introduce the functions $x = x(\lambda, \phi)$ and $y = y(\lambda, \phi)$ and use them as Cartesian coordinates on a flat rectangular piece of paper. Each choice of the two functions corresponds to a different *map projection*. The *Mercator projection* is defined by

$$x = \frac{W\phi}{2\pi}, \quad y = \frac{H}{2\pi} \ln \left[\tan \left(\frac{\pi}{4} + \frac{\lambda}{2} \right) \right],$$

where W and H are the width and height of the map respectively. Find the line element for this projection.

- 2.6 For the general map projection discussed in Exercise 1.5, show that the angle between two directions at some point on the Earth's surface will equal the angle between the corresponding directions on the map, provided that the functions x and y are chosen such that

$$\Omega(x, y)(dx^2 + dy^2) = a^2(d\lambda^2 + \cos^2 \lambda d\phi^2),$$

for some function $\Omega(x, y)$. Show that the Mercator projection satisfies this condition. Write down the general requirement on x and y for an *equal-area* projection, in which the area of any region of the map is proportional to the corresponding area on the Earth's surface. Find such a projection. Is it possible to obtain a projection that simultaneously is equal-area and preserves angles?

- 2.7 A *conformal transformation* is not a change of coordinates but an actual change in the geometry of a manifold such that the metric tensor transforms as

$$\tilde{g}_{ab}(x) = \Omega^2(x)g_{ab}(x),$$

where $\Omega(x)$ is some non-vanishing scalar function of position. In a pseudo-Riemannian manifold, show that if $x^a(\lambda)$ is a null curve with respect to g_{ab} (i.e. $ds^2 = 0$ along the curve), then it is also a null curve with respect to $\tilde{g}_{ab}$. Is this true for timelike curves?

- 2.8 A curve on the surface of a 2-sphere of radius a is defined parametrically by $\theta = u$, $\phi = 2u - \pi$, where $0 \leq u \leq \pi$. Sketch the curve and show that its total length is

$$L = a \int_0^\pi \sqrt{1 + 4 \sin^2 u} du.$$

Show that, in general, the length of a curve is independent of the parameter used to describe it.

- 2.9 Show that the line element of a 3-sphere of radius a embedded in four-dimensional Euclidean space can be written in the form

$$ds^2 = a^2[d\chi^2 + \sin^2 \chi(d\theta^2 + \sin^2 \theta d\phi^2)].$$

Hence, in this three-dimensional non-Euclidean space, calculate the area of the 2-sphere defined by $\chi = \chi_0$. Also find the total volume of the three-dimensional space.

- 2.10 Consider the three-dimensional space with line element

$$ds^2 = \frac{dr^2}{1 - 2\mu/r} + r^2(d\theta^2 + \sin^2 \theta d\phi^2),$$

and calculate the following quantities:

- the area of a sphere of coordinate radius $r = R$;
- the 3-volume of a sphere of coordinate radius $r = R$;
- the radial distance between the sphere $r = 2\mu$ and the sphere $r = 3\mu$;
- the 3-volume contained between the two spheres in part (c).

Verify that your answers reduce to the usual Euclidean results in the limit $\mu \rightarrow 0$.

- 2.11 Prove the following results used in Section 2.11:

- $(\partial x^a / \partial x'^b)_p$ has N^2 independent values;
- $(\partial^2 x^a / \partial x'^b \partial x'^c)_p$ has $\frac{1}{2}N^2(N+1)$ independent values;
- $(\partial^3 x^a / \partial x'^b \partial x'^c \partial x'^d)_p$ has $\frac{1}{6}N^2(N+1)(N+2)$ independent values;
- $g'_{ab}(P)$ has $\frac{1}{2}N(N+1)$ independent values;
- $(\partial g'_{ab} / \partial x'^c)_p$ has $\frac{1}{2}N^2(N+1)$ independent values;
- $(\partial^2 g'_{ab} / \partial x'^c \partial x'^d)_p$ has $\frac{1}{4}N^2(N+1)^2$ independent values.

Hence show that, in a general Riemannian manifold, at least $N^2(N^2 - 1)/12$ of the second derivatives $(\partial^2 g'_{ab} / \partial x'^c \partial x'^d)_p$ will not vanish in any coordinate system.

- 2.12 Consider the two-dimensional space with line element

$$ds^2 = \frac{dr^2}{1 - 2\mu/r} + r^2 d\phi^2.$$

Using the result (2.25), show that this geometry can be embedded in three-dimensional Euclidean space, and find the equations for the corresponding two-dimensional surface.

- 2.13 By identifying a suitable coordinate transformation, show that the line element

$$ds^2 = (c^2 - a^2 t^2) dt^2 - 2at dt dx - dx^2 - dy^2 - dz^2,$$

where a is a constant, can be reduced to the Minkowski line element.